Cassiopée Project: Optimal Transport

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Table des matières

1. Problem Context **2**

2. Problem Formulation and Theoretical Foundations **2**
A. Counting Measures . . . 2
B. Monge Problem : Discrete Case . . . 2
C. Monge Problem : Continuous Case . . . 6
D. Kantorovich Problem : Discrete Case . . . 7
E. Kantorovich Problem : Continuous Case . . . 9
F. Comparison between Monge and Kantorovich . . . 10
G. Wasserstein Distance : Discrete Case . . . 10
H. Wasserstein Distance : Continuous and Special Cases, Geometric Properties . . . 11
I. Dual Problem . . . 13

3. Entropic Regularization **16**
A. Reminders . . . 16
B. Regularized Kantorovich Problem : Discrete Case . . . 17
C. Regularized Kantorovich Problem : Continuous Case . . . 19
D. Sinkhorn Algorithm . . . 20

4. Algorithmic Foundations **26**
A. Complementarity of the Primal and Dual Problems . . . 26
B. Graphs and Algorithms . . . 26

5. Conclusion **28**

6. References **29**

1. Problem Context

The mathematician Gaspard Monge proposed the following problem in 1781 : suppose we have n coal mines that extract ores and n factories that require them to operate. To transport a mass m_{ij} of ore from a mine i to a **single** factory j, there is a cost per unit of transported mass c_{ij} that depends on the transport distance. Does an optimal mine-factory assignment exist that minimizes the total cost of transporting ore from mines to factories ? We can generalize Monge's problem to the Kantorovich problem as the search for a minimum-cost mine-factory assignment where the mass at i may be delivered to **multiple** factories. The solutions to these problems have numerous applications in machine learning.

2. Problem Formulation and Theoretical Foundations

A. Counting Measures

Let us revisit the terms of the problem in detail. We would first like to model the map of mine locations : let $\mathcal{X}$ be the set of mine positions. A mine $i \in [\![1, n]\!]$ is at a position $x_i \in \mathcal{X}$ and holds a normalized mass of ore to be transported $a_i \in [0, 1]$.

Definition 2.A.1. (Histogram) Let us define the set of histograms. A histogram is also called a probability vector :

$$\Sigma_n = \left\{ a \in \mathbb{R}^n : \sum_{i=1}^{n} a_i = 1 \right\} \tag{1}$$

We can then interpret the total masses to be transported with a counting measure at the points $(x_1, \ldots, x_n) \in \mathcal{X}^n$ with respective weights $(a_1, \ldots, a_n) \in \Sigma_n$.

Definition 2.A.2. (Counting Measure) We can now construct our probability measure :

$$\alpha = \sum_{i=1}^{n} a_i \delta_{x_i} \text{ where } \delta_{x_i} \text{ is a Dirac mass at position } x_i. \tag{2}$$

Property 2.A.3. For any f continuous on $\mathcal{X}$ with a counting measure α :

$$\int_{\mathcal{X}} f(x)\, d\alpha(x) = \int_{\mathcal{X}} f(x) \sum_{i=1}^{n} a_i \delta_{x_i}(dx) = \sum_{i=1}^{n} a_i \int_{\mathcal{X}} f(x) \delta_{x_i}(dx) = \sum_{i=1}^{n} a_i f(x_i) \tag{3}$$

Similarly, we define $\mathcal{Y}$ as the set of factory positions. A factory $j \in [\![1, m]\!]$ is at a position $y_j \in \mathcal{Y}$ and requires a normalized mass of ore $b_j \in [0, 1]$ to operate. We can then interpret the total masses received with a counting measure at the points $(y_1, \ldots, y_n) \in \mathcal{Y}^m$ with respective weights $(b_1, \ldots, b_m) \in \Sigma_m$.

B. Monge Problem : Discrete Case

Consider the following two counting measures α and β :

$$\alpha = \sum_{i=1}^{n} a_i \delta_{x_i} \quad \beta = \sum_{i=1}^{m} b_i \delta_{y_i} \text{ where } \begin{cases} (x_1, \ldots, x_n) \in \mathcal{X}^n \text{ and } (a_1, \ldots, a_n) \in \Sigma_n \\ (y_1, \ldots, y_m) \in \mathcal{Y}^m \text{ and } (b_1, \ldots, b_m) \in \Sigma_m \end{cases} \tag{4}$$

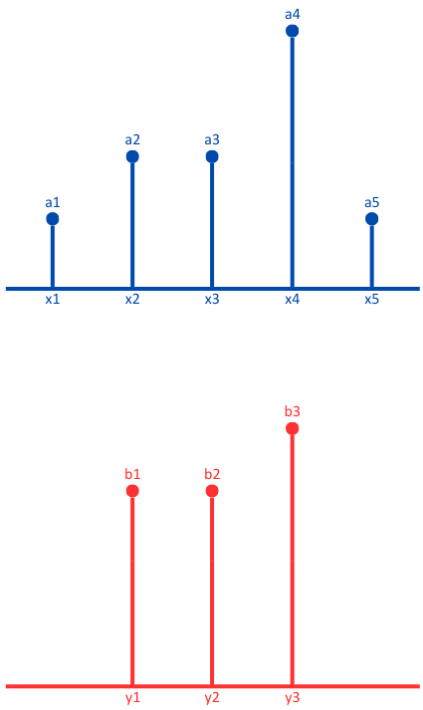

FIGURE 1 – Examples of counting measures. The weights of the blue measure represent the quantities of ore to be transported, while the weights of the red measure represent the ore requirements. The objective is to minimize the transport cost between these two sets of measures.

Definition 2.B.1. (Transport Map) The Monge problem consists in finding a map that associates each x_i to a y_j and moves the mass from x_i to y_j with the lowest possible total displacement cost. Such a map $T : \{x_1, \dots, x_n\} \to \{y_1, \dots, y_n\}$ is a solution to the Monge problem and is called a transport map. A transport map must satisfy the following mass conservation criterion :

$$\forall j \in [\![1, m]\!] \quad b_j = \sum_{i:T(x_i)=y_j} a_i \tag{5}$$

In other words, the mass b_j must equal the sum of the masses coming from the preimage of y_j. It is also implied that the departing masses cannot split, like snails each carrying their shell to their destination. A factory may have several mines as preimages.

Definition 2.B.2. (Mass Conservation) We can write the mass conservation constraint (5) of the transport map T between α and β in a more compact form :

$$T_{\#}\alpha = \beta \tag{6}$$

Definition 2.B.3. (Discrete Monge Problem) The Monge transport cost minimization problem is written in the form :

$$\min_T \left\{ \sum_i c(x_i, T(x_i)) : T_{\#}\alpha = \beta \right\} \tag{7}$$

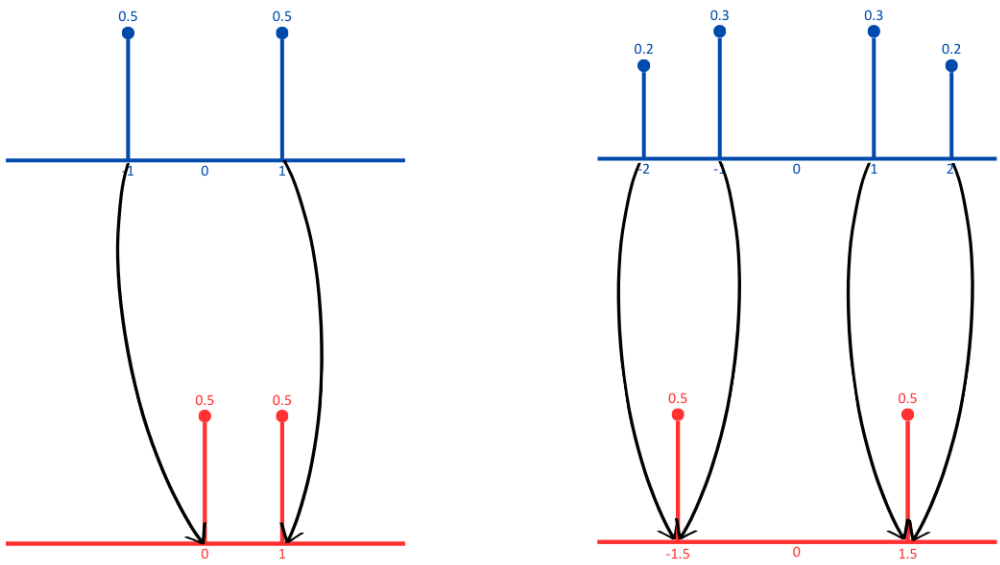

FIGURE 2 – Examples of transport maps. On the left, we define T a transport map such that T(-1) = 0 and $T(1) = 1$. T satisfies criterion (5) : the weights do not split. On the right, we define another transport map such that T(-2) = T(-1) = -1.5 and $T(1) = T(2) = 1.5$. Again, T satisfies (5) : the masses do not split and are added together.

Where $c(x_i, T(x_i))$ is the cost of transporting the mass at x_i to $T(x_i)$. The left-hand term is the sum of the costs of each transport, the right-hand term represents the mass conservation of the map T between α and β. The solution to the discrete Monge problem is therefore the transport map T that minimizes criterion (7

Remark 2.B.4. (Figure 3) If $n = m$ and if all masses are equal to $\frac{1}{n}$, the mass conservation constraint implies that the Monge problem reduces to finding a permutation of the indices of the support points of the source density with the smallest cost.

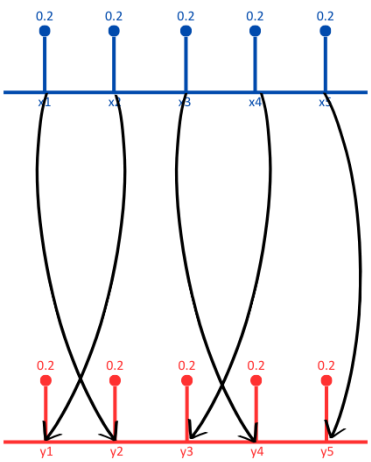

FIGURE 3 – We have defined a transport map T such that $T(x_1) = y_2$, $T(x_2) = y_1$, $T(x_3) = y_4$, $T(x_4) = y_3$ and $T(x_5) = y_5$. T is a permutation of $[\![1; 5]\!]$. We observe that the support contracts : there are fewer positions at the destination than at the source !

Remark 2.B.5. (Figure 4) If $n \neq m$, the solution to the Monge problem may not exist. This occurs when the source and destination mass vectors are incompatible, which is always the case when $n < m$.

Remark 2.B.6. We note that the cardinality of the image of a transport map is smaller than that of its domain because masses can be added together, and therefore the support points of the image

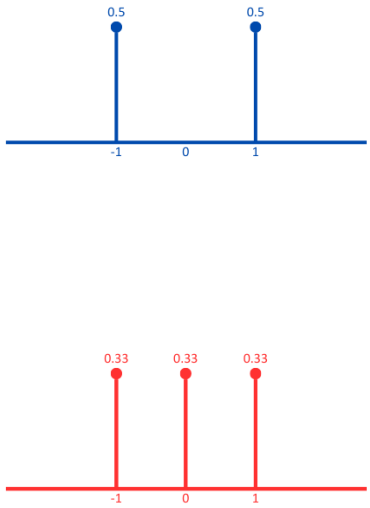

FIGURE 4 – No transport map is possible from the first to the second measure without mass splitting.

can overlap.

We see that when a transport map between two distributions is possible, the support positions of the source measure with their associated masses are displaced to obtain the target measure. Therefore...

Definition 2.B.7. (Push Forward) For a transport map $T : \mathcal{X} \to \mathcal{Y}$, (6) can be interpreted as an operator $T_\# : \mathcal{M}(\mathcal{X}) \to \mathcal{M}(\mathcal{Y})$ called the push forward, which moves the support positions of the source measure with their associated masses. In particular, we can move several points to the same location so that their associated masses are added. More formally :

$$T_\#\alpha = \sum_{i=1}^{n} a_i \delta_{T(x_i)} \tag{8}$$

Remark 2.B.8. Let $B \subseteq \mathcal{Y}$. To measure the mass of B, we can measure the mass of the preimage of B under $T_\#$. We then sense that we can measure the target set by a change of variable which is $T_\#$.

Let us formalize our intuition in the case of the continuous Monge problem !

C. Monge Problem : Continuous Case

We no longer consider two discrete counting measures α and β but two continuous measures α and β satisfying (6). The Monge problem then consists in finding a map that associates each $x_i \in \mathrm{Supp}(\alpha)$ to a $y_j \in \mathrm{Supp}(\beta)$ and moves the infinitesimal-width strip at x_i to y_j with the lowest possible transport cost (see figure 5).

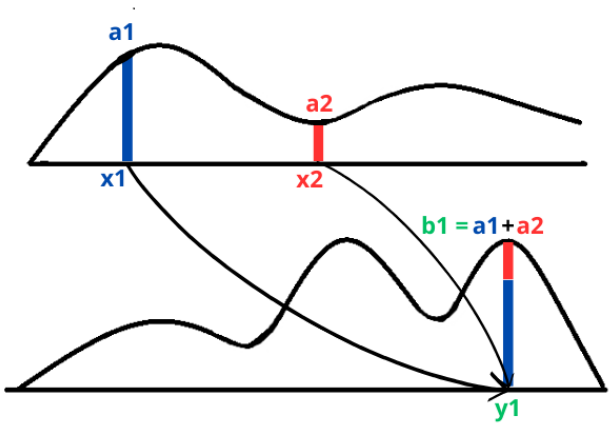

FIGURE 5 – We move the infinitesimal strips of x_1 and x_2 to y_1. The two strips are added together.

Remark 2.C.1. (Measure via Preimage) Let $(E, \mathcal{E})$ and $(F, \mathcal{F})$ be two measurable spaces. Let μ be a measure on $(E, \mathcal{E})$, and let X be a random variable from E to F. For any $B \subseteq F$, we can define a measure $X_{\#}\mu$ on the space $(F, \mathcal{F})$:

$$X_{\#}\mu(A) = \mu(X^{-1}(B)) \tag{9}$$

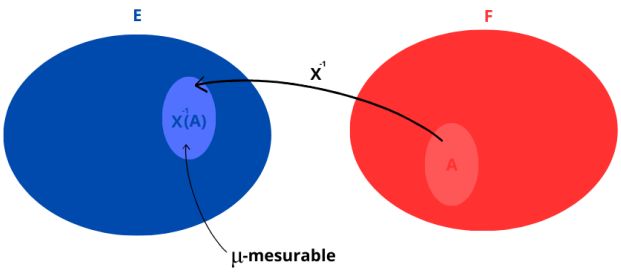

FIGURE 6 – Measuring a set via its preimage.

Definition 2.C.2. (Push Forward) Keeping in mind remarks 2.B.8. and 2.C.1., for $T : \mathcal{X} \to \mathcal{Y}$, the image measure $\beta = T_{\#}\alpha \in \mathcal{M}(\mathcal{Y})$ of a measure $\alpha \in \mathcal{M}(\mathcal{X})$ satisfies :

$$\forall h \in \mathcal{C}(\mathcal{Y}), \quad \int_{\mathcal{Y}} h(y)\, d\beta(y) = \int_{\mathcal{X}} h(T(x))\, d\alpha(x). \tag{10}$$

Equivalently, for any measurable set $B \subset \mathcal{Y}$, we have :

$$\beta(B) = \alpha(\{x \in \mathcal{X} : T(x) \in B\}) = \alpha(T^{-1}(B)). \tag{11}$$

Definition 2.C.3. (Continuous Monge Problem) (7) then becomes in the continuous case.

$$\min_T \left\{ \int_X c(x, T(x))\, d\alpha(x) : T_{\#}\alpha = \beta \right\}. \tag{12}$$

The solution to the continuous Monge problem is therefore the transport map T that minimizes criterion (12).

D. Kantorovich Problem : Discrete Case

In 1942, Leonid Kantorovich proposed another formulation of Monge's problem without being aware of Monge's work. He proposed a more general formulation of the problem by allowing masses to split, meaning they can travel to multiple destinations. We now need to find a mine-factory assignment that respects the criterion of the following paragraph.

Definition 2.D.1. Let $(a, b) \in \Sigma_n \times \Sigma_m$ be the respective source and target histograms. Let us then define the set $U(a, b)$ such that :

$$U(a, b) = \{M \in \mathbb{R}_+^{n \times m} \text{ such that } M\mathbb{1}_n = a \text{ and } M\mathbb{1}_m^T = b\}. \tag{13}$$

This is the set of mass transport matrices that respect mass conservation between a and b. The difference from the previous formulations is that there are no longer constraints on n and m : the source masses can split to travel to distinct destinations !

Remark 2.D.2. $U(a, b)$ is closed. Indeed, let $(P_k)_{k \in \mathbb{N}} \in U(a, b)^{\mathbb{N}}$ be a convergent sequence. For all $i \in [\![1, n]\!]$, and for all $k \in \mathbb{N}$, $(P_k)_{i,1} + \cdots + (P_k)_{i,n} = a_i$. Taking the limit, $\lim_{k\to\infty}(P_k)_{i,1} + \cdots + \lim_{k\to\infty}(P_k)_{i,n} = a_i$. Proceeding similarly for the second constraint of $U(a, b)$, $\lim_{k\to\infty} P_k \in U(a, b)$, so $U(a, b)$ is closed by sequential characterization !

Example 2.D.3. (Figure 7) We wish to move the blue discrete measure to the red discrete measure.

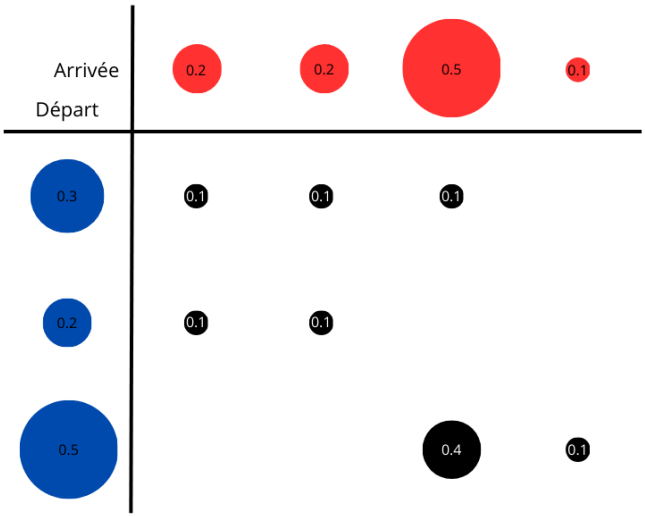

FIGURE 7 – Example to illustrate the discrete Kantorovich problem.

Let us then define the histograms a and b representing the blue and red measures respectively.

$$a = \begin{pmatrix} 0.3 \\ 0.2 \\ 0.5 \end{pmatrix} \quad b = \begin{pmatrix} 0.2 \\ 0.2 \\ 0.5 \\ 0.1 \end{pmatrix}$$

Then :

$$M = \begin{bmatrix} 0.1 & 0.1 & 0.1 & 0 \\ 0.1 & 0.1 & 0 & 0 \\ 0 & 0 & 0.4 & 0.1 \end{bmatrix} \in U(a, b)$$

For all $(i, j) \in [\![1, 3]\!] \times [\![1, 4]\!]$, M_{ij} represents the quantity of mass going from the ith position of the source measure to the jth mass of the target measure. In particular, the sum of the jth column corresponds to all the masses arriving at the jth position of the target measure. Similarly, the sum of the ith row corresponds to all the masses coming from the ith position of the source measure. M therefore belongs to $U(a, b)$.

Definition 2.D.4. (Discrete Kantorovich Problem and Transport Plan) For a given cost matrix C. The Kantorovich transport cost minimization problem is written in the form :

$$L_C(a, b) = \min_{M \in U(a,b)} \langle C, M \rangle = \min_{M \in U(a,b)} \sum_{i,j} c_{i,j} M_{i,j} \tag{14}$$

For all $(i, j) \in [\![1, n]\!] \times [\![1, m]\!]$, $M_{i,j}$ represents the mass transported from i to j, and $c_{i,j}$ the unit transport cost from the ith position of the source measure to the jth position of the target measure. The solution to the discrete Kantorovich problem is therefore a matrix M that minimizes (14). M is called a transport plan.

Remark 2.D.5. We note that M is a joint distribution whose marginals are fixed and correspond to the source/blue and target/red measures.

Example 2.D.6. (Figure 8) Let us return to the Kantorovich problem where we examine a transport between two uniform distributions $a = \frac{1}{3}\mathbb{1}_3$ and $b = \frac{1}{3}\mathbb{1}_3$. We then have the following mass matrix for all $(i, j) \in [\![1, 3]\!]^2$.

$$M_{i,j} = \begin{cases} \frac{1}{n} & \text{if } i = \sigma(j) \\ 0 & \text{otherwise} \end{cases}$$

The mass matrix M is then :

$$M = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & 0 & \frac{1}{3} \\ 0 & \frac{1}{3} & 0 \end{bmatrix}$$

Remark 2.D.7. According to Birkhoff's theorem, the optimal solution to L_C is a permutation matrix. Thus, instead of searching for the solution among the set of 3×3 matrices, we search for it among the set of permutation matrices.

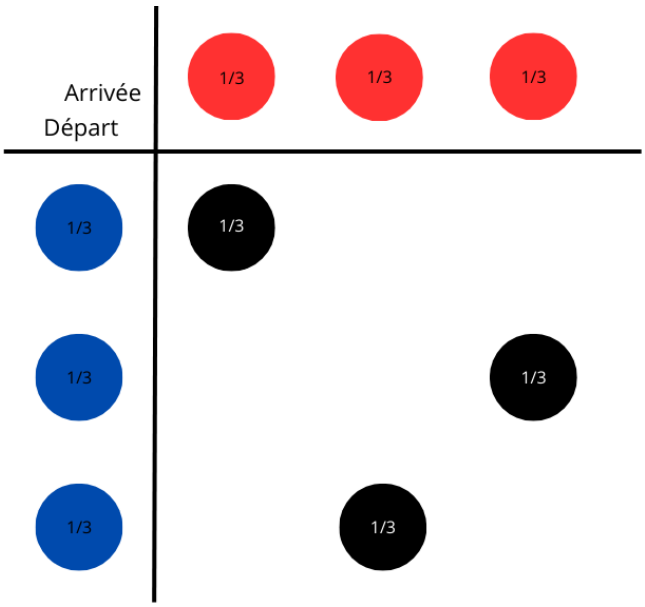

FIGURE 8 – Example to illustrate the discrete Kantorovich problem.

$$L_C\left(\frac{1}{n}\mathbb{1}_n, \frac{1}{n}\mathbb{1}_n\right) \leq \min_{\sigma \in \mathrm{Perm}(n)} \langle C, M_\sigma \rangle \tag{15}$$

E. Kantorovich Problem : Continuous Case

We no longer consider two histograms a and b, but two measures α and β. We then need a mass conservation criterion between α and β. We therefore transform $U(a,b)$ into $\mathcal{U}(\alpha,\beta)$ and then $L_C(a,b)$ into $\mathcal{L}_C(\alpha,\beta)$: in summary, we move from the discrete to the continuous case.

Definition 2.E.1. (Set of Measures to Transport) According to remark 2.D.5., the mine-factory assignment we seek is a joint distribution π whose marginals we fix as $\mathcal{P}_{\mathcal{X}\#}\pi = \alpha$ and $\mathcal{P}_{\mathcal{Y}\#}\pi = \beta$. By fixing the marginals, the mass conservation criterion is satisfied. We therefore seek π in the set :

$$\mathcal{U}(\alpha,\beta) = \left\{\pi \in \mathcal{M}_+^1(\mathcal{X}\times\mathcal{Y}) : \mathcal{P}_{\mathcal{X}\#}\pi = \alpha \text{ and } \mathcal{P}_{\mathcal{Y}\#}\pi = \beta\right\} \tag{16}$$

Definition 2.E.2. (Continuous Kantorovich Problem) Returning to (14), $c(x,y)$ is analogous to c_{ij} and $\pi(x,y)$ is analogous to P_{ij}. The integral is analogous to the sum. Each infinitesimal displacement at (x,y) is associated with a cost $\mathrm{c}(x,y)$.

$$\mathcal{L}_C(\alpha,\beta) = \min_{\pi \in \mathcal{U}(\alpha,\beta)} \int_{\mathcal{X}\times\mathcal{Y}} c(x,y)d\pi(x,y) \tag{17}$$

F. Comparison between Monge and Kantorovich

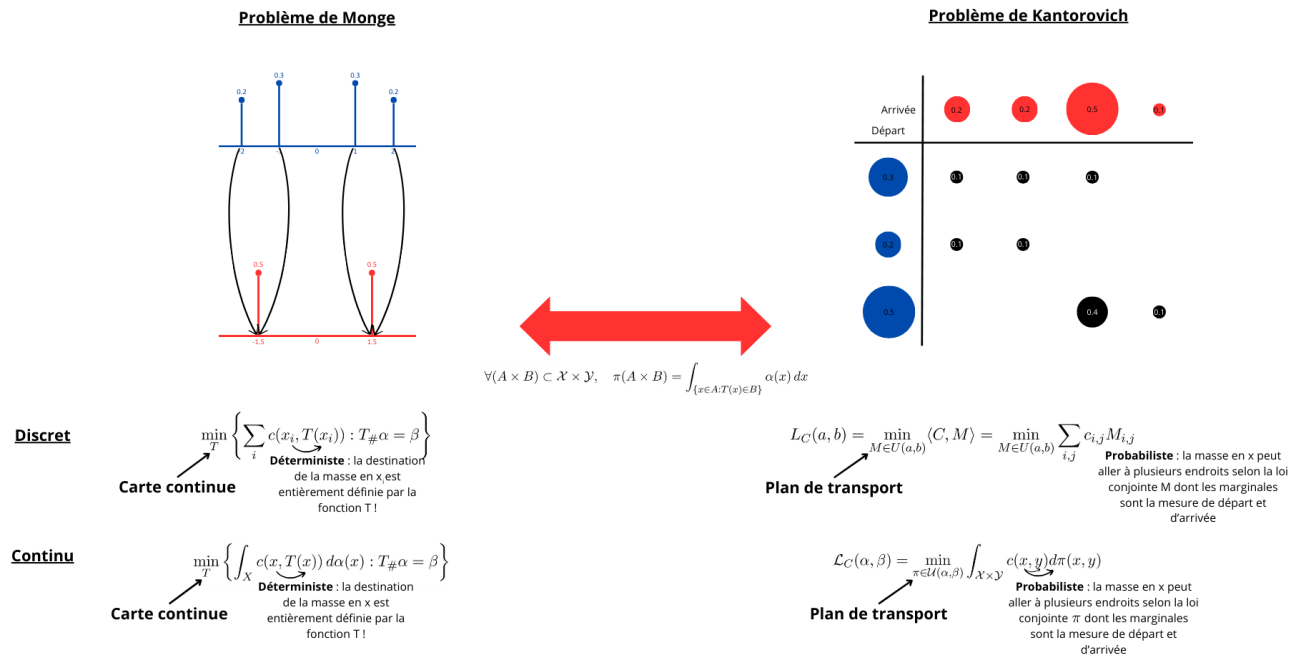

FIGURE 9 – Comparative table between the Monge problem and the Kantorovich problem. The Monge problem is described as deterministic because to compute the transport cost, the destination of each mass must be fully determined by the function T. The Kantorovich problem is described as probabilistic because masses can split toward multiple locations given by the joint distribution.

G. Wasserstein Distance : Discrete Case

An important mathematical object for understanding the rest of this guide is the Wasserstein distance. In the discrete case, the Wasserstein distance, also known as the optimal transport distance or Kantorovich distance, is a measure that quantifies the "distance" between two histograms. We will then extend this measure to quantify the distance between two continuous probability measures. In other words, if Ω is a bounded subspace of $\mathbb{R}^d$, $P(\Omega)$ is the set of probability measures whose supports are in Ω, then $P(\Omega)$ equipped with the Wasserstein distance is a metric space.

Definition 2.F.1. (Distance Matrix) $\mathbf{D} \in \mathbb{R}_+^{n \times n}$ is a distance matrix on $[\![n]\!]$ if and only if the following three properties are satisfied :

1. $\mathbf{D} \in \mathbb{R}_+^{n \times n}$ is symmetric ;
2. $\mathbf{D}_{i,j} = 0$ iff $i = j$;
3. $\forall (i,j,k) \in [\![n]\!]^3$, $\mathbf{D}_{i,k} \leq \mathbf{D}_{i,j} + \mathbf{D}_{j,k}$.

Definition 2.F.2. (Wasserstein Distance between Two Histograms) For all $(a,b) \in \Sigma_n \times \Sigma_m$ the respective source and target histograms, for all $p \geq 1$, and for any distance matrix $\mathbf{D} \in \mathbb{R}_+^{n \times m}$, we define the Wasserstein distance between a and b by :

$$W_p(a,b) = \left(L_{\mathbf{D}_p}(a,b) \right)^{\frac{1}{p}} \tag{18}$$

Proof 2.F.3.

1. **Symmetry :**

$$\begin{aligned}
W_p(a,b) &= \min_{P\in U(a,b)} \sum_{i,j} c_{i,j} P_{i,j} \\
&= \min_{P\in U(a,b)} \sum_{i,j} D^p_{i,j} P_{i,j} \\
&= \min_{P^T\in U(b,a)} \sum_{i,j} D^p_{j,i} P_{j,i} \\
&= \min_{P^T\in U(b,a)} \sum_{i,j} (D^p_{i,j})^T P^T_{i,j} \\
&= W_p(b,a).
\end{aligned}$$

2. **Positive definiteness :**
3. **Triangle inequality :** Let : $\tilde{b} \in \mathbb{R}^n$ such that

$$\tilde{b}_i = \begin{cases} b_i & \text{if } b_i > 0, \\ 1 & \text{if } b_i = 0. \end{cases}$$

For $P \in U(a,b)$ and $Q \in U(b,c)$, let us also set :

$$S = P\text{diag}\left(\frac{1}{\tilde{b}}\right) Q.$$

We then note that $S \in U(a,c)$ because :

$$S\mathbb{1}_n = P\text{diag}\left(\frac{1}{\tilde{b}}\right) Q\mathbb{1}_n = P\left(\frac{b}{\tilde{b}}\right) = P\mathbb{1}_{\text{Supp}(b)} = a$$

and similarly $S^T\mathbb{1}_n = c$.

It then follows that :

$$\begin{aligned}
W_p(a,c) &= \min_{P\in U(a,c)} \langle P, D^p\rangle^{\frac{1}{p}} \\
&\leq \langle S, D^p\rangle^{\frac{1}{p}} \quad \text{since } S \in U(a,c) \\
&= \left(\sum_{ik} D^p_{ik} \sum_j \frac{P_{ij}Q_{jk}}{\tilde{b}_j}\right)^{\frac{1}{p}} \\
&\leq \left(\sum_{ijk} (D_{ij} + D_{jk})^p \sum_j \frac{P_{ij}Q_{jk}}{\tilde{b}_j}\right)^{\frac{1}{p}} \quad \text{(Triangle inequality)} \\
&\leq \left(\sum_{ijk} D^p_{ij} \frac{P_{ij}Q_{jk}}{\tilde{b}_j}\right)^{\frac{1}{p}} + \left(\sum_{ijk} D^p_{jk} \frac{P_{ij}Q_{jk}}{\tilde{b}_j}\right)^{\frac{1}{p}} \quad \text{(Minkowski inequality)} \\
&= W_p(a,b) + W_p(b,c).
\end{aligned}$$

H. Wasserstein Distance : Continuous and Special Cases, Geometric Properties

Let Ω be a bounded subspace of $\mathbb{R}^d$. For example, if $d = 1$, we can choose $\Omega = [0,1]$, a signal subspace, or if $d = 2$, we can choose $\Omega = [0,1]^2$, an image subspace. Let $P(\Omega)$ be the set of probability densities on Ω.

Definition 2.H.1. (Wasserstein Distance between Two Measures) The p-Wasserstein metric denoted $\mathcal{W}_p$ is defined below :

$$\forall p \geq 1, \quad \forall (I_0, I_1) \in P(\Omega)^2, \quad \mathcal{W}_p(I_0, I_1) = \left(\inf_{\gamma \in \mathcal{U}(I_0, I_1)} \int_{\Omega \times \Omega} |x-y|^p \, d\gamma(x,y) \right)^{\frac{1}{p}} \tag{19}$$

This formula is analogous to equation (18) where the transport cost from x to y is $|x-y|^p$:

$$W_p(a,b) = \left(L_{\mathbf{D}_p}(a,b)\right)^{\frac{1}{p}} \quad \text{becomes} \quad \mathcal{W}_p(\alpha, \beta) = (\mathcal{L}_C(\alpha, \beta))^{\frac{1}{p}} \tag{20}$$

Property 2.H.2. For all $p > 1$, $(\mathcal{P}(\Omega), \mathcal{W}_p)$ is called the p-Wasserstein space and is a metric space.

Property 2.H.3. For all $n \in \mathbb{N}$:

$$\left(W_p(I_n, I) \underset{n\to\infty}{\to} 0\right) \iff \left(\forall f : \Omega \to \mathbb{R} \text{ continuous and bounded}, \quad \int_\Omega f(x) I_n(x)\, dx \underset{n\to\infty}{\to} \int_\Omega f(x) I(x)\, dx\right) \tag{21}$$

Vocabulary 2.H.4. If $p = 1$, $\mathcal{W}_p$ is called the Monge-Rubinstein metric.

Definition 2.H.5. (Cumulative Distribution Function) For all $x \in \mathbb{R}$, we can compute the cumulative distribution function of the density I

$$F_I^{\leftarrow}(x) = \int_{\inf(\Omega)}^{x} I(t)\, dt \tag{22}$$

Definition 2.H.6. (Pseudo-inverse $F_I^{\leftarrow}$ of the Cumulative Distribution Function) For all $y \in [0,1]$, we can compute the pseudo-inverse function

$$F_I^{\leftarrow}(y) = \inf\{x \in \Omega \mid F_I(x) > y\} \tag{23}$$

Property 2.H.7. If F_I is an increasing function, then the inverse of F_I is the pseudo-inverse $F_I^{\leftarrow}$. Moreover, in the case where $d = 1$, we have an explicit expression of the Wasserstein distance. For all $(I_0, I_1) \in \mathcal{P}(\Omega)^2$:

$$W_p(I_0, I_1) = \left(\int_0^1 \left| F_0^{-1}(z) - F_1^{-1}(z) \right|^p \, dz \right)^{\frac{1}{p}} \tag{24}$$

We now want to find a transport between two high-dimensional probability distributions. For example, $\Omega \subset \mathbb{R}^2$ and not $\Omega \subset \mathbb{R}$. The idea is therefore to project the distributions to obtain 1D distributions, then use the Wasserstein distance (explicit in 1D) and then superpose the projections. To project we use the Radon transform, which amounts to integrating an image along a line.

Recall 2.H.9. (IMA 4101, Radon Transform) For all $I \in L^1(\mathbb{R}^2) = \{I : \mathbb{R}^2 \to \mathbb{R}$ such that $\int_{\mathbb{R}^2} |I(x)|\, dx < \infty\}$ and for all $(\theta, t) \in \mathbb{R} \times \mathbb{S}^1$ (where $\mathbb{S}^1$ is the unit sphere in $\mathbb{R}^2$),

$$\mathcal{R}I(t, \theta) = \int_{\mathbb{R}} I(t\theta + s\theta^{\perp})\, ds \tag{25}$$

More generally, for all $I \in L^1(\mathbb{R}^d) = \{I : \mathbb{R}^d \to \mathbb{R}$ such that $\int_{\mathbb{R}} |I(x)|\, dx < \infty\}$ and for all $(\theta, t) \in \mathbb{R} \times \mathbb{S}^{d-1}$ (where $\mathbb{S}^{d-1}$ is the unit sphere in $\mathbb{R}^d$),

$$\mathcal{R}I(t, \theta) = \int_{\mathbb{R}} I(t\theta + s\theta^{\perp})\, ds \tag{26}$$

Therefore,

$$\begin{cases} \mathcal{R} : L^1(\mathbb{R}^d) \to L(\mathbb{R} \times S^{(d-1)}) \\ L^1(\mathbb{R} \times S^{(d-1)}) \text{ is an infinite set of 1D densities} \end{cases} \tag{27}$$

Definition 2.H.10. (Superposition of Projections, Sliced Wasserstein Distance) Sliced Wasserstein metric for PDFs I_0 and I_1 on $\mathbb{R}^d$:

$$\begin{cases} \int_{\mathbb{R}^d} \mathcal{W}_p(I_0, I_1) = \left(\int_{S^{d-1}} \mathcal{W}_p(\mathcal{R}(I_0)(.,\theta), \mathcal{R}(I_1)(.,\theta))\, d\theta \right)^{\frac{1}{p}} \\ \text{with } \mathcal{R}I_0(.,\theta) \in L^1(\mathbb{R} \times \theta) \text{ is a 1D density since } \theta \text{ is fixed} \end{cases} \tag{28}$$

Definition 2.H.11. (Path Length) In a metric space (X, d), the length of a path $I : [a, b] \to X$ is defined by :

$$\text{Length of } I = L(I) = \sup \left\{ \sum_{m=1}^{n} d(I(t_{m-1}), I(t_m)) \mid a = t_0 < t_1 < \cdots < t_n = b \right\} \tag{29}$$

Definition 2.H.12. (Geodesic Space) A metric space (X, d) is a geodesic space if for all I_0 and I_1, there exists a path $I : [0, 1] \to X$ such that $I(0) = I_0$ and $I(1) = I_1$, and for all $0 \leq s \leq t \leq 1$ where $d(I(s), I(t)) = L(I|_{[s,t]})$. In particular, the length of I equals the distance from $I(0)$ to $I(1)$.

Property 2.H.13. Let Ω be convex, $(P(\Omega), W_p)$ is a geodesic space.

I. Dual Problem

Let us adopt another perspective on the Monge-Kantorovich problem. Suppose an operator seeks to move ore to factories at minimum cost but decides to outsource this task to a logistics provider. This provider applies a two-part pricing scheme : a price f_i for collecting one unit of resource at each mine i, and a price g_j for delivering one unit of resource to each factory j. The provider seeks to maximize their revenue while ensuring that the prices f and g satisfy the inequality $f_i + g_j \leq c_{ij}$, where c_{ij} is the direct transport cost of a unit mass of ore from mine i to factory j. If the operator verifies that these inequalities are satisfied, they can conclude that the provider's offer is advantageous.

Indeed, the provider's optimal solution for prices f and g is equivalent to the minimum cost the operator would obtain by solving the transport problem themselves. This dual formulation illustrates an effective approach for verifying the competitiveness of outsourcing offers and has practical implications for reducing computational costs.

Recall 2.I.1. (Discrete Kantorovich Problem, from a Linear Programming Perspective)
Let a so-called cost matrix : $C \in \mathbb{R}^{n \times m}$ as well as another matrix $P \in \mathbb{R}^{n \times m}$. We denote $\langle C, P \rangle = \sum_{i=1}^{n} \sum_{j=1}^{m} C_{i,j} P_{i,j}$. The Kantorovich optimal transport problem between 2 discrete measures : $a = \sum_{i=1}^{n} a_i \delta_{x_i}$ and $b = \sum_{j=1}^{m} b_j \delta_{y_j}$ consists in determining $P \in \mathbb{R}^{n \times m}$ solution to the following linear program (P_1) :

$$(P_1) \quad \begin{cases} \min & \langle C, P \rangle + \epsilon \mathrm{KL}(P \| a \otimes b) \\ \text{subject to :} & \sum_{j=1}^{m} P_{i,j} = a_i, \quad i = 1, \ldots, n \\ & \sum_{i=1}^{n} P_{i,j} = b_j, \quad j = 1, \ldots, m \\ & P_{i,j} \geq 0, \quad (i,j) \in \{1, \ldots, n\} \times \{1, \ldots, m\} \end{cases} \tag{30}$$

Property 2.I.2. (Convex Kantorovich Optimization Problem) Let us define the functions :

$$\begin{cases} \forall i \in \{1,\dots,n\} \quad h_i : \mathbb{R}^{n\times m} \to \mathbb{R} \quad P \mapsto h_i(P) = a_i - \sum_{j=1}^{m} P_{i,j} = a_i - \langle [P]_{i,:}, 1_m \rangle \\ \forall j \in \{1,\dots,m\} \quad g_j : \mathbb{R}^{n\times m} \to \mathbb{R} \quad P \mapsto g_j(P) = b_j - \sum_{i=1}^{n} P_{i,j} = b_j - \langle [P]_{:,j}, 1_n \rangle \end{cases}$$

Where for $P \in \mathbb{R}^{n\times m}$, $[P]_{i,:}$ and $[P]_{:,j}$ denote respectively the i-th row and j-th column of P. Finally, let us define :

$$\forall (i,j) \in \{1,\dots,n\} \times \{1,\dots,m\} \quad f_{i,j} : \begin{cases} \mathbb{R}^{n\times m} \to \mathbb{R} \\ P \mapsto f_{i,j}(P) = -P_{i,j} \end{cases}$$

We observe that the functions h_i and g_j are affine, as are the functions $f_{i,j}$. Consequently, they are convex. We are therefore dealing with a convex optimization problem (P_2) :

$$(P_2) \quad \begin{cases} \min & \langle C, P \rangle \\ \text{subject to :} & h_i(P_2) = 0 \quad \forall i \in \{1,\dots,n\} \\ & g_j(P_2) = 0 \quad \forall j \in \{1,\dots,m\} \\ & f_{i,j}(P_2) \le 0 \quad \forall (i,j) \in \{1,\dots,n\} \times \{1,\dots,m\} \end{cases} \tag{31}$$

For all $(P, \lambda, (\alpha, \beta)) \in \mathbb{R}^{n\times m} \times \mathbb{R}^{n\times m} \times \mathbb{R}^{n+m}$, the Lagrangian associated with (P_2) is then :

$$\begin{aligned} L(P,\lambda,(\alpha,\beta)) &= f(P) + \sum_{1\le i\le n, 1\le j\le m} \lambda_{i,j} f_{i,j}(P) + \sum_{i=1}^{n} \alpha_i h_i(P) + \sum_{j=1}^{m} \beta_j g_j(P) \\ &= \langle C, P\rangle - \langle \lambda, P\rangle + \sum_{i=1}^{n} \alpha_i (a_i - \langle [P]_{i,:}, 1_m\rangle) + \sum_{j=1}^{m} \beta_j (b_j - \langle [P]_{:,j}, 1_n\rangle) \\ &= \langle C-\lambda, P\rangle + \langle \alpha, a\rangle - \langle \sum_{i=1}^{n} \alpha_i [P]_{i,:}, 1_m\rangle + \langle \beta, b\rangle - \langle \sum_{j=1}^{m} \beta_j [P]_{:,j}, 1_n\rangle \\ &= \langle C-\lambda, P\rangle + \langle \alpha, a\rangle - \langle P^T\alpha, 1_m\rangle + \langle \beta, b\rangle - \langle P\beta, 1_n\rangle \\ &= \langle C-\lambda, P\rangle + \langle \alpha, a\rangle - \langle (P^T\alpha)^T, 1_m^T\rangle + \langle \beta, b\rangle - \langle (P\beta)^T, 1_n^T\rangle \\ &= \langle C-\lambda, P\rangle + \langle \alpha, a\rangle - 1_m\alpha^T P + \langle \beta, b\rangle - P\beta 1_m^T \\ &= \langle C-\lambda, P\rangle + \langle \alpha, a\rangle - \langle \alpha 1_m^T, P\rangle + \langle \beta, b\rangle - \langle \beta^T 1_n, P\rangle \\ &= \langle C-\lambda-\alpha 1_m^T - \beta^T 1_n, P\rangle + \langle \alpha, a\rangle + \langle \beta, b\rangle \end{aligned}$$

Thus, the dual function φ associated with the convex problem (P_2) is :

$$\begin{aligned} \varphi(\lambda,(\alpha,\beta)) &= \inf_{P\in\mathbb{R}^{n\times m}} L(P,\lambda,(\alpha,\beta)) \\ &= \inf_{P\in\mathbb{R}^{n\times m}} \langle C-\lambda-\alpha 1_m^T - \beta^T 1_n, P\rangle + \langle \alpha, a\rangle + \langle \beta, b\rangle \\ &= \begin{cases} \langle \alpha, a\rangle + \langle \beta, b\rangle & \text{if } C-\lambda-\alpha 1_m^T - \beta^T 1_n = 0 \\ -\infty & \text{otherwise} \end{cases} \end{aligned}$$

Property 2.I.3. (Dual Problem of the Convex Kantorovich Optimization Problem) Finally, the dual problem (P_2') associated with our convex optimization problem is the following :

$$(P_2') \quad \begin{cases} \max & \langle \alpha, a \rangle + \langle \beta, b \rangle \\ \text{subject to :} & \lambda \geq 0 \\ & C - \lambda - \alpha 1_m^T - \beta^T 1_n = 0 \end{cases} \tag{32}$$

$$\iff \quad (P_2') \quad \begin{cases} \max & \langle \alpha, a \rangle + \langle \beta, b \rangle \\ \text{subject to :} & C - \alpha 1_m^T - \beta^T 1_n \geq 0 \end{cases} \tag{33}$$

3. Entropic Regularization

The Kantorovich optimal transport problem consists in finding an optimal way to transport a mass distribution to another, by minimizing a certain transport cost. However, the direct solution to this problem is often difficult to compute numerically due to the complex and sometimes discontinuous nature of optimal solutions. To overcome these difficulties, a commonly used technique consists in adding an entropic regularization term to the original problem. Thus, solving the new problem provides an approximation of the solution to the original problem at lower computational cost. Entropic regularization has several advantages :

— **Simplicity of minimization :** The minimization of the regularized problem can be carried out using simple alternating minimization schemes. These schemes translate into iterations in the form of matrix-vector products, which is particularly well-suited to GPU computations.
— **Reduced memory requirements :** For certain applications, matrix-vector products do not require storing a cost matrix of size $n \times m$, but only access to a kernel evaluation, which considerably reduces memory requirements.
— **Computational speedup :** In the case where a large group of measures shares the same support, these matrix-vector products can be reformulated as matrix-matrix products, enabling significant computational speedups.
— **Smoothness of the solution :** The approximate distance obtained by regularization is smooth with respect to the input histogram weights and the Dirac positions, enabling efficient automatic differentiation.

A. Reminders

Definition 3.A.1. (Coupling Matrix with Fixed Marginal Distributions) Consider a matrix P in the set of admissible couplings $U(a, b)$, that is :

$$U(a, b) = \{P \in \mathbb{R}_+^{n \times m} \text{ such that } P\mathbb{1}_n = a \text{ and } P\mathbb{1}_m^T = b\}. \tag{34}$$

The matrix P must satisfy :

$$\sum_j P_{i,j} = a_i \quad \text{and} \quad \sum_i P_{i,j} = b_j \tag{35}$$

Definition 3.A.2. (Entropy of a Coupling Matrix) For $(a, b) \in \Sigma_n \times \Sigma_m$ the respective source and target histograms, the discrete entropy of a coupling matrix $P \in U(a, b)$ is defined by :

$$H(P) \coloneqq -\sum_{i,j} P_{i,j}(\log(P_{i,j}) - 1), \tag{36}$$

with an analogous definition for vectors, with the convention that $H(a) = -\infty$ if any of the entries a_i is zero or negative. To maximize this entropy subject to the margin constraints, we must find the most "uniform" distribution P possible.

Definition 3.A.3. (Tensor Product $a \otimes b$) For $(a, b) \in \Sigma_n \times \Sigma_m$, the tensor product $a \otimes b$ is given by :

$$(a \otimes b)_{i,j} = a_i b_j. \tag{37}$$

Remark 3.A.4. $a \otimes b \in U(a, b)$, indeed :

$$\sum_j (a \otimes b)_{i,j} = \sum_j a_i b_j = a_i \sum_j b_j = a_i \cdot 1 = a_i \tag{38}$$

$$\sum_i (a \otimes b)_{j,i} = \sum_i a_i b_j = b_j \sum_i a_i = b_j \cdot 1 = b_j \tag{39}$$

Definition 3.A.5. (Kullback-Leibler Divergence) The Kullback-Leibler divergence between two couplings is defined as :

$$\mathrm{KL}(P\|K) := \sum_{i,j} P_{i,j} \left(\log \left(\frac{P_{i,j}}{K_{i,j}} \right) - P_{i,j} + K_{i,j} \right) \tag{40}$$

B. Regularized Kantorovich Problem : Discrete Case

Definition 3.B.1. (Regularized Kantorovich Problem in the Discrete Case) Using the notation of (14), the optimal transport problem regularized by an entropic term can be formulated in the discrete case as follows :

$$L_C^\epsilon(a,b) = \min_{P \in U(a,b)} \langle C, P \rangle - \epsilon H(P) = \min_{P \in U(a,b)} \langle C, P \rangle + \epsilon \sum_{i,j} P_{i,j}(\log(P_{i,j}) - 1) \tag{41}$$

where $\langle C, P \rangle$ is the linear transport cost between the distributions, $H(P)$ is the entropy of the transport matrix P, $U(a,b)$ represents the set of admissible transport matrices that satisfy the given margin constraints, and $\epsilon > 0$ is the regularization parameter.

Remark 3.B.2. The impact of ϵ is twofold :

1. **Small values of** ϵ : For small values of ϵ, the solution P_ϵ tends to approach the solution of the classical optimal transport problem without regularization. In this regime, the cost term $\langle C, P \rangle$ predominates, leading to a solution that primarily minimizes the transport cost without significantly considering the dispersion of P.
2. **Large values of** ϵ : Conversely, for large values of ϵ, the entropic term $-\epsilon H(P)$ becomes dominant. This favors smoother and more dispersed solutions, away from the extreme solutions of the unregularized transport problem. The solution P_ϵ tends to be more uniform, minimizing the impact of specific transport costs in favor of a more homogeneous distribution of flows.

The effect of ϵ is therefore to balance between fidelity to the transport cost and the smoothness of the solution. In practice, choosing an appropriate ϵ is crucial to achieve a desired trade-off between these two aspects, depending on the specific needs of the application considered. This choice can be guided by practical considerations such as the numerical stability of the solving algorithm and the nature of the transport data.

Property 3.B.3. (Convergence with respect to ϵ) The regularized Kantorovich problem (41) admits a unique solution P_ϵ and it converges toward the solution of the Kantorovich problem (14) with the greatest entropy among the set of optimal solutions. More formally :

$$P_\epsilon \xrightarrow[\epsilon \to 0]{} \arg \min_{P \in U(a,b)} \{-H(P) : \langle C, P \rangle = L_C(a,b)\} \tag{42}$$

In particular :

$$\begin{cases} L_C^\epsilon(a,b) \xrightarrow[\epsilon \to 0]{} L_C(a,b), \\ P_\epsilon \xrightarrow[\epsilon \to +\infty]{} a \otimes b = ab^T = (a_i b_j)_{i,j}. \end{cases} \tag{43}$$

Proof 3.B.4. Let $(\epsilon_\ell)_\ell$ be a sequence such that $\epsilon_\ell \to 0$ and $\epsilon_\ell > 0$. We denote P_{ϵ_ℓ} the solution to (41) for $\epsilon = \epsilon_\ell$. Since $U(a,b)$ is bounded, we can extract a subsequence (which we will not rename without loss of generality so as not to impede the readability of the proof) such that $P_{\epsilon_\ell} \to P^*$. Since $U(a,b)$ is closed, $P^* \in U(a,b)$. We consider P such that $\langle C, P\rangle = L_C(a,b)$. By optimality of P and P_{ϵ_ℓ} for their respective optimization problems (for $\epsilon = 0$ and $\epsilon = \epsilon_\ell$), we know that each P_{ϵ_ℓ} is optimal for its specific regularization problem, which implies that it minimizes the modified total cost, $\langle C, P\rangle - \epsilon_\ell H(P)$, better than any other admissible transport matrix, including a matrix P optimal for the unregularized problem ($\epsilon = 0$). Therefore, by comparing the modified cost of P_{ϵ_ℓ} to that of P, we obtain :

$$\langle C, P_{\epsilon_\ell}\rangle - \epsilon_\ell H(P_{\epsilon_\ell}) \leq \langle C, P\rangle - \epsilon_\ell H(P) \tag{44}$$

This inequality indicates that the difference between the cost transported by P_{ϵ_ℓ} and that transported by P, adjusted for the effect of regularization, is always positive. It also shows how regularization affects the optimization of the transport cost.

$$0 \leq \langle C, P_{\epsilon_\ell}\rangle - \langle C, P\rangle \leq \epsilon_\ell(H(P_{\epsilon_\ell}) - H(P)). \tag{45}$$

Since H is continuous, taking the limit $\ell \to +\infty$ in this expression shows that $\langle C, P^*\rangle = \langle C, P\rangle$, so P^* is a feasible solution to (41). Moreover, dividing by ϵ_ℓ in (44) and taking the limit shows that $H(P) \leq H(P^*)$, which shows that P^* is a solution to (41). Since the solution P_0^* of this program is unique by the strict convexity of $-H$, we have $P^* = P_0^*$.

Proof 3.B.4. (continued) (Special case $\epsilon \to +\infty$) We have seen that P_ϵ is defined as the solution to (30) :

$$\min_{P \in U(a,b)} \langle C, P\rangle - \epsilon H(P) \tag{46}$$

$$\iff \quad \min_{P \in U(a,b)} \langle C, P\rangle + \epsilon KL(P \| a \otimes b) \tag{47}$$

When $\epsilon \to +\infty$, the term $+\epsilon H(P)$ becomes dominant. Thus, the optimization problem is equivalent to :

$$\max_{P \in U(a,b)} H(P) \tag{48}$$

$$\iff \quad \min_{P \in U(a,b)} KL(P \| a \otimes b) \tag{49}$$

To show that $a \otimes b$ is the solution to (41), let us compute the entropy of this matrix. For each element $(i,j) \in [\![1,n]\!] \times [\![1,m]\!]$, we have :

$$(a \otimes b)_{i,j} \log \frac{(a \otimes b)_{i,j}}{(a \otimes b)_{i,j}} = (a \otimes b)_{i,j} \log 1 = 0$$

Consequently, by summing over all elements i, or by integrating over the whole space if the distributions are continuous, the KL divergence of any distribution with respect to itself is always equal to zero :

$$\mathrm{KL}(a \otimes b \| a \otimes b) = 0 \tag{50}$$

We want to show that when $\epsilon \to +\infty$, the solution P_ϵ converges to $a \otimes b$. Let us evaluate the energy at $a \otimes b$ (which belongs to the constraint set $U(a, b)$), we have :

$$\langle C, P_\epsilon \rangle + \epsilon\, \mathrm{KL}(P_\epsilon \| a \otimes b) \leq \langle C, a \otimes b \rangle + \epsilon\, \mathrm{KL}(a \otimes b \| a \otimes b) \tag{51}$$

and since $\langle C, P_\epsilon \rangle \geq 0$, this leads to

$$\mathrm{KL}(P_\epsilon \| a \otimes b) \leq \epsilon^{-1} \langle C, a \otimes b \rangle \leq \frac{\|C\|_\infty}{\epsilon} \tag{52}$$

since all norms are equivalent (as $\mathbb{R}^{n \times m}$ is finite-dimensional), we can choose any of them. We choose to take $\| \cdot \|_\infty$ where for all $P \in \mathbb{R}^{n \times m}$, $\|P\|_\infty = \max_{1 \leq i \leq n, 1 \leq j \leq m} |P_{ij}|$. Thus, $\mathrm{KL}(P_\epsilon \| a \otimes b) \to 0$ and therefore $P_\epsilon \to a \otimes b$ since KL is a valid norm-divergence. Thus, we have shown that the matrix $a \otimes b = (a_i b_j)_{i,j}$ satisfies the margin constraints and maximizes entropy among all admissible coupling matrices, which explains why $P_\epsilon \xrightarrow[\epsilon \to +\infty]{} a \otimes b$.

Remark 3.B.5. For all $(a, b) \in \Sigma_n \times \Sigma_m$:

$$\begin{aligned}
H(a \otimes b) &= -\sum_{i,j} (a_i b_j) \left(\log(a_i b_j) - 1 \right) \\
&= -\sum_{i,j} a_i b_j (\log a_i + \log b_j - 1) \\
&= -\sum_{i,j} a_i b_j \log a_i - \sum_{i,j} a_i b_j \log b_j + \sum_{i,j} a_i b_j \\
&= -\sum_i a_i \log a_i - \sum_j b_j \log b_j + 1 \quad \text{where } \sum_i a_i = 1 \text{ and } \sum_j b_j = 1. \\
&= H(a) + H(b) + 1
\end{aligned}$$

C. Regularized Kantorovich Problem : Continuous Case

Definition 3.B.5. (Continuous Regularized Model) The regularized optimal transport problem (30) can be extended to the continuous case. Using the notation of (17) :

$$\min_{\pi \in \mathcal{U}(\alpha, \beta)} \left\{ \int_{\mathcal{X} \times \mathcal{Y}} c(x, y)\, d\pi(x, y) + \varepsilon \int_{\mathcal{X} \times \mathcal{Y}} \pi(x, y) \log \pi(x, y)\, dx\, dy \right\} \tag{53}$$

where $\mathcal{U}(\alpha, \beta)$ represents the set of admissible transport plans between the distributions α and β, $c(x, y)$ is the transport cost between points x and y, and $\varepsilon > 0$ is the regularization parameter.

D. Sinkhorn Algorithm

Definition 3.D.1. (Regularized Kantorovich Problem with KL, Continuous Case) We can consider arbitrary measures by replacing the discrete entropy with the relative entropy with respect to the product of measures $\mathrm{d}\alpha \otimes \mathrm{d}\beta(x,y) = \mathrm{d}\alpha(x)\mathrm{d}\beta(y)$, and propose a regularized counterpart of the problem with the Kullback-Leibler divergence :

$$L^{\xi}(\alpha,\beta) := \min_{\pi \in U(\alpha,\beta)} \int_{X\times Y} c(x,y)\mathrm{d}\pi(x,y) + \epsilon \mathrm{KL}(\pi \| \alpha \otimes \beta) \tag{54}$$

where the relative entropy is a generalization of the discrete Kullback-Leibler divergence :

$$\mathrm{KL}(\pi\|\xi) = \int_{X\times Y} \log\left(\frac{\mathrm{d}\pi}{\mathrm{d}\xi}(x,y)\right)\mathrm{d}\pi(x,y) + \int_{X\times Y} (\mathrm{d}\xi(x,y) - \mathrm{d}\pi(x,y)) \tag{55}$$

and by convention $\mathrm{KL}(\pi\|\xi) = +\infty$ if π does not have a density with respect to ξ. It is important to realize that the reference measure $\alpha \otimes \beta$ chosen to define the regularization term $\mathrm{KL}(\pi\|\alpha\otimes\beta)$ plays no specific role (we explain why subsequently), only its support matters.

Interpretation 3.D.2. (Thermodynamics) This problem is intended to model the most probable coupling between gas particles that can only be observed at two different moments. The parameter ϵ controls the temperature of the gas, while particles move in a deterministic straight line as in optimal transport for the Euclidean cost or according to a probabilistic Brownian bridge.

Interpretation 3.D.3. (Probabilistic) If $(X,Y) \sim \pi$ have marginals $X \sim \alpha$ and $Y \sim \beta$, then $\mathrm{KL}(\pi\|\alpha\otimes\beta) = I(X,Y)$ is the mutual information of the pair, which is zero if and only if X and Y are independent. The entropic problem (54) is then equivalent to :

$$\min_{(X,Y)\sim\pi, X\sim\alpha, Y\sim\beta} E(c(X,Y)) + \epsilon I(X,Y) \tag{56}$$

Having introduced a reformulation of the Kantorovich problem, we explore here an algorithm based on this reformulation that converges toward the optimal solution of our optimization problem. The objective is to construct a sequence $(P_n)_{n\geq 1}$ converging toward a P^*, the optimal solution to our initial problem. Intuitively, each P_n represents the unique solution to a modified optimization problem, increasingly resembling the original problem as n increases, thanks to the addition of a continuous and strongly convex function controlled by $\frac{1}{n}$. The function chosen to ensure this property is the Kullback-Leibler divergence, which guarantees the uniqueness of the solution at each step.

Recall 3.D.4. (Kullback-Leibler Divergence) Let $P, Q \in [0,1]^{n\times m}$ such that $\sum_{i=1}^n \sum_{j=1}^m P_{ij} = \sum_{i=1}^n \sum_{j=1}^m Q_{ij} = 1$. The Kullback-Leibler divergence between P and Q, denoted $\mathrm{KL}(P\|Q)$, is defined as follows :

$$\mathrm{KL}(P\|Q) = \sum_{i,j} P_{ij} \log\left(\frac{P_{ij}}{Q_{ij}}\right) - P_{ij} + Q_{ij} \tag{57}$$

If a term is of the form "0 log(0)", that term is considered to be zero. If $Q_{ij} = 0$ and $P_{ij} \neq 0$ for some (i,j), then $\mathrm{KL}(P\|Q) = \infty$.

Properties 3.D.5. Let $P \in \mathbb{R}_+^{n\times m}$ and $Q \in \mathbb{R}_+^{n\times m}$ such that $\sum_{i=1}^n \sum_{j=1}^m P_{ij} = \sum_{i=1}^n \sum_{j=1}^m Q_{ij} = 1$. The following properties hold :

1. $\mathrm{KL}(P\|Q) \geq 0$.
2. $\mathrm{KL}(P\|Q) = 0$ if and only if $P = Q$.
3. The function $P \mapsto \mathrm{KL}(P\|Q)$ is 1-strongly convex on the set of probability distributions that have the same support as Q.

Proof 3.D.6.

1. We have :
$$\mathrm{KL}(P\|Q) = \sum_{i,j} P_{ij} \log\left(\frac{P_{ij}}{Q_{ij}}\right) = -\sum_{i,j} P_{ij} \log\left(\frac{Q_{ij}}{P_{ij}}\right) \tag{58}$$
Now, we know that $P_{ij} \in [0,1]$ and $\sum_{i,j} P_{ij} = 1$. Moreover, the log function is concave. Thus, the function $-\log$ is convex and we have :
$$\sum_{i,j} P_{ij} \log\left(\frac{P_{ij}}{Q_{ij}}\right) \leq \log\left(\sum_{i,j} \frac{P_{ij}}{Q_{ij}} P_{ij}\right) \leq \log\left(\sum_{i,j} Q_{ij}\right) \tag{59}$$
Therefore, $\mathrm{KL}(P\|Q) \geq 0$.
2. Suppose $\mathrm{KL}(P\|Q) = 0$. Then :
$$\sum_{i,j} Q_{ij} \log\left(\frac{P_{ij}}{Q_{ij}}\right) = 0 = \mathbb{E}_Q\left[\log\left(\frac{P}{Q}\right)\right] = \log(\mathbb{E}_Q\left[\frac{P}{Q}\right]) = \log\left(\sum_{i,j} \frac{P_{ij}}{P_{ij}Q_{ij}}\right) \tag{60}$$
We are therefore in the equality case of Jensen's inequality. So necessarily, there exists $c \in \mathbb{R}$ such that $\frac{P}{Q} = c$. Thus, for all $(i,j) \in \{1,\dots,n\} \times \{1,\dots,m\}$, we have :
$$\frac{P_{ij}}{Q_{ij}} = c \tag{61}$$
therefore,
$$\sum_{i,j} P_{ij} = c \sum_{i,j} Q_{ij} \tag{62}$$
Since $\sum_{i,j} P_{ij} = \sum_{i,j} Q_{ij} = 1$, it follows that $c = 1$. In conclusion, for all $(i,j) \in \{1,\dots,n\} \times \{1,\dots,m\}$, we have :
$$P_{ij} = Q_{ij} \tag{63}$$
Thus, we have $P = Q$.
3. The function $\mathrm{KL}(\cdot \parallel Q)$ is C^∞ on the set of probability distributions sharing the same support as Q. For all $(i,j) \in \{1,\dots,n\} \times \{1,\dots,m\}$,
$$\frac{\partial \mathrm{KL}(P \parallel Q)}{\partial P_{ij}} = \ln\left(\frac{P_{ij}}{Q_{ij}}\right) + 1 - 1 + 0 = \ln\left(\frac{P_{ij}}{Q_{ij}}\right). \tag{64}$$
Differentiating again with respect to $P_{i',j'}$:
$$\frac{\partial^2 \mathrm{KL}(P \parallel Q)}{\partial P_{i',j'} \partial P_{ij}} = \begin{cases} \frac{1}{P_{ij}} & \text{if } (i,j) = (i',j'), \\ 0 & \text{otherwise.} \end{cases} \tag{65}$$
Consequently, the Hessian of $\mathrm{KL}(P \parallel Q)$ is a matrix in $\mathbb{R}^{n\times m}$ with diagonal elements equal to $\frac{1}{P_{ij}} \geq 1$, proving that $\mathrm{KL}(\cdot \parallel Q)$ is indeed 1-strongly convex.

The Kullback-Leibler divergence is defined by :
$$\mathrm{KL}(P\|Q) = \sum_{i,j} \varphi\left(\frac{P_{ij}}{Q_{ij}}\right) Q_{ij}, \tag{66}$$

where $\varphi(s) = s\log(s)$. For any convex function φ such that $\varphi(1) = 0$, we have, by Jensen's inequality :

$$\sum_{i,j} \varphi\left(\frac{P_{ij}}{Q_{ij}}\right) Q_{ij} \geq \varphi\left(\sum_{i,j} \frac{P_{ij}}{Q_{ij}} Q_{ij}\right). \tag{67}$$

(Optimization Problem and Hessians) For our new optimization problems denoted $L_C^{1/n}(a,b)$ where P_n are unique solutions, we consider the Kullback-Leibler divergence controlled by $\frac{1}{n}$. Choosing the right reference measure is crucial for practical management, for example, to easily know its marginal distributions. We select $Q = a \otimes b$ where $(a \otimes b)_{ij} = a_i b_j$ for all $(i,j) \in \{1,\ldots,n\} \times \{1,\ldots,m\}$.

Proposition 3.D.7. Given $P \in U(a,b)$ and $(c,d) \in \Sigma_n \times \Sigma_m$ discrete probabilities with weights $c_1,\ldots,c_n$ and $d_1,\ldots,d_m$ respectively :

$$\mathrm{KL}(P \parallel a \otimes b) = \mathrm{KL}(P \parallel c \otimes d) - \mathrm{KL}(a \parallel c) - \mathrm{KL}(b \parallel d) \tag{68}$$

Proof 3.D.8. The proof follows directly from the properties of logarithms and the linearity of expectation.

We have :

$$\begin{aligned}
\mathrm{KL}(P \parallel c \otimes d) - \mathrm{KL}(P \parallel a \otimes b) &= \sum_{i,j} P_{ij} \log\left(\frac{P_{ij}}{c_i d_j}\right) - \sum_{i,j} P_{ij} \log\left(\frac{P_{ij}}{a_i b_j}\right) \\
&= \sum_{i,j} P_{ij}[\log(P_{ij}) - \log(c_i d_j)] - \sum_{i,j} P_{ij}[\log(P_{ij}) - \log(a_i b_j)] \\
&= \sum_{i,j} P_{ij} \log\left(\frac{a_i b_j}{c_i d_j}\right) \\
&= \sum_{i,j} P_{ij} \log\left(\frac{a_i}{c_i}\right) + \sum_{i,j} P_{ij} \log\left(\frac{b_j}{d_j}\right) \\
&= \sum_{i} \left(\log\left(\frac{a_i}{c_i}\right) \sum_j P_{ij}\right) + \sum_{j} \left(\log\left(\frac{b_j}{d_j}\right) \sum_i P_{ij}\right) \\
&= \sum_{i} a_i \log\left(\frac{a_i}{c_i}\right) + \sum_{j} b_j \log\left(\frac{b_j}{d_j}\right) \text{ since } P \in U(a,b) \\
&= \mathrm{KL}(a \parallel c) + \mathrm{KL}(b \parallel d)
\end{aligned}$$

Let us then pose a discrete version of the KL-regularized Kantorovich problem (54) :

$$L_C^{\frac{1}{n}}(a,b) \stackrel{\text{def}}{=} \min_{P \in U(a,b)} \left(\langle P, C\rangle + \frac{1}{n}\mathrm{KL}(P \parallel a \otimes b)\right) \tag{69}$$

Proposition 3.D.9. (Continuous Generalization of (68)) For all $\pi \in U(\alpha,\beta)$, and for all (α',β') having the same measure sets as (α,β) (so that they both have densities with respect to each other), we have :

$$\mathrm{KL}(\pi \| \alpha \otimes \beta) = \mathrm{KL}(\pi \| \alpha' \otimes \beta') - \mathrm{KL}(\alpha \otimes \beta \| \alpha' \otimes \beta') \tag{70}$$

Proposition 3.D.10. Let $P \in U(a,b)$, it is the unique solution to the problem $L_C^\epsilon(a,b)$ if and only if there exist $(u,v) \in \mathbb{R}_+^n \times \mathbb{R}_+^m$ such that :

$$P = \text{diag}(u) K \text{diag}(v) \tag{71}$$

Moreover, our P will be in $U(a,b)$ if and only if it satisfies : $u \odot (Kv) = a$ and $v \odot (K^T u) = b$.

Proof 3.D.11. The optimization problem we are trying to solve is the following :

$$(P_1) \quad \begin{cases} \min\limits_P & \langle C, P \rangle + \epsilon \text{KL}(P \| a \otimes b) \\ \text{subject to :} & h_i(P) = 0 \quad \forall i \in \{1, \dots, n\}, \\ & g_j(P) = 0 \quad \forall j \in \{1, \dots, m\}, \\ & f_{i,j} \leq 0 \quad \forall (i,j) \in \{1, \dots, n\} \times \{1, \dots, m\}, \end{cases} \tag{72}$$

where the h_i, g_j and $f_{i,j}$ are exactly the same as those in the dual problem. Moreover, following the same reasoning as for the dual problem (we simply replace $\langle C, P \rangle$ by $\langle C, P \rangle + \epsilon \text{KL}(P \| a \otimes b)$), the Lagrangian associated with this problem will be :

$$L(P, \lambda, \alpha, \beta) =< C, P > + \epsilon \text{KL}(P \| a \otimes b) + (\alpha, a - 1_m^T P) + (\beta, b - P 1_n) \tag{73}$$

where α, β are Lagrange multipliers associated with the constraints. This can be expressed more explicitly as :

$$L(P, \lambda, \alpha, \beta) = \sum_{i,j} C_{ij} P_{ij} + \epsilon P_{ij} \ln\left(\frac{P_{ij}}{a_i b_j}\right) + \sum_i \alpha_i a_i + \sum_j \beta_j b_j - \sum_{i,j} \alpha_i P_{ij} - \sum_{i,j} \beta_j P_{ij} \tag{74}$$

Since the problem is convex and satisfies Slater's conditions, our P satisfies the KKT conditions, in particular the stationarity condition. Therefore : $\nabla_P L(P, \alpha, \beta) = 0$.

$$\begin{aligned} \frac{\partial L}{\partial P_{ij}} = 0 &\Rightarrow C_{ij} + \epsilon \left(\frac{\partial}{\partial P_{ij}} P_{ij} \ln\left(\frac{P_{ij}}{a_i b_j}\right)\right) - \alpha_i - \beta_j = 0 \\ &\Rightarrow C_{ij} + \epsilon \ln\left(\frac{P_{ij}}{a_i b_j}\right) - \alpha_i - \beta_j = 0 \\ &\Rightarrow \ln\left(\frac{P_{ij}}{a_i b_j}\right) = \frac{\alpha_i + \beta_j - C_{ij}}{\epsilon} \\ &\Rightarrow P_{ij} = a_i b_j \exp\left(\frac{\alpha_i + \beta_j - C_{ij}}{\epsilon}\right) \\ &\Rightarrow P_{ij} = a_i e^{\frac{\alpha_i}{\epsilon}} e^{-\frac{C_{ij}}{\epsilon}} b_j e^{\frac{\beta_j}{\epsilon}} \\ &\Rightarrow P_{ij} = u_i K_{ij} v_j, \quad \text{where } u_i = a_i e^{\frac{\alpha_i}{\epsilon}}, \quad K_{ij} = e^{-\frac{C_{ij}}{\epsilon}}, \quad \text{and } v_j = e^{\frac{\beta_j}{\epsilon}} \\ &\Rightarrow P = \text{diag}(u) K \text{diag}(v) \end{aligned}$$

Thus, for this P, we have :

$$P \in U(a,b) \iff P \text{diag}(u) K \text{diag}(v) = a \text{ and } P \text{diag}(u) K^T \text{diag}(v) = b \tag{75}$$

We are therefore no longer searching for an entire matrix but for two vectors : u and v. But since these u and v satisfy : $u \odot (Kv) = a$ and $v \odot (K^T u) = b$. The Sinkhorn algorithm proposes to solve this iteratively, where we initialize u to the vector 1_m, allowing us to obtain u via the relation $u \odot (Kv) = a$ and then v via the other relation.

Algorithm 1 Sinkhorn Algorithm

0: **Input :**
0: $K \in \mathbb{R}_+^{n \times m}$: the matrix defined above.
0: $a \in \mathbb{R}_+^n$: vector of target row sums
0: $b \in \mathbb{R}_+^m$: vector of target column sums
0: $\epsilon > 0$: small positive number for convergence
0: max_iter : maximum number of iterations
0: **Output :**
0: $P \in \mathbb{R}_+^{n \times m}$: adjusted transport matrix
0: **Begin**
0: Initialize $u \leftarrow 1_m$ (vector of size n with all elements equal to 1)
0: **for** $t = 1$ **to** max_iter **do**
0: Update u :
0: $u \leftarrow \frac{a}{Kv}$
0: Update v :
0: $v \leftarrow \frac{b}{K^T u}$
0: Check convergence :
0: **if** the difference between the u and v vectors from the previous and current iterations is less than ϵ **then**
0: Break out of the loop
0: **end if**
0: **end for**
0: Compute the transport matrix P :
0: $P \leftarrow \text{diag}(u) K \text{diag}(v)$
0: Return P
0: **End** =0

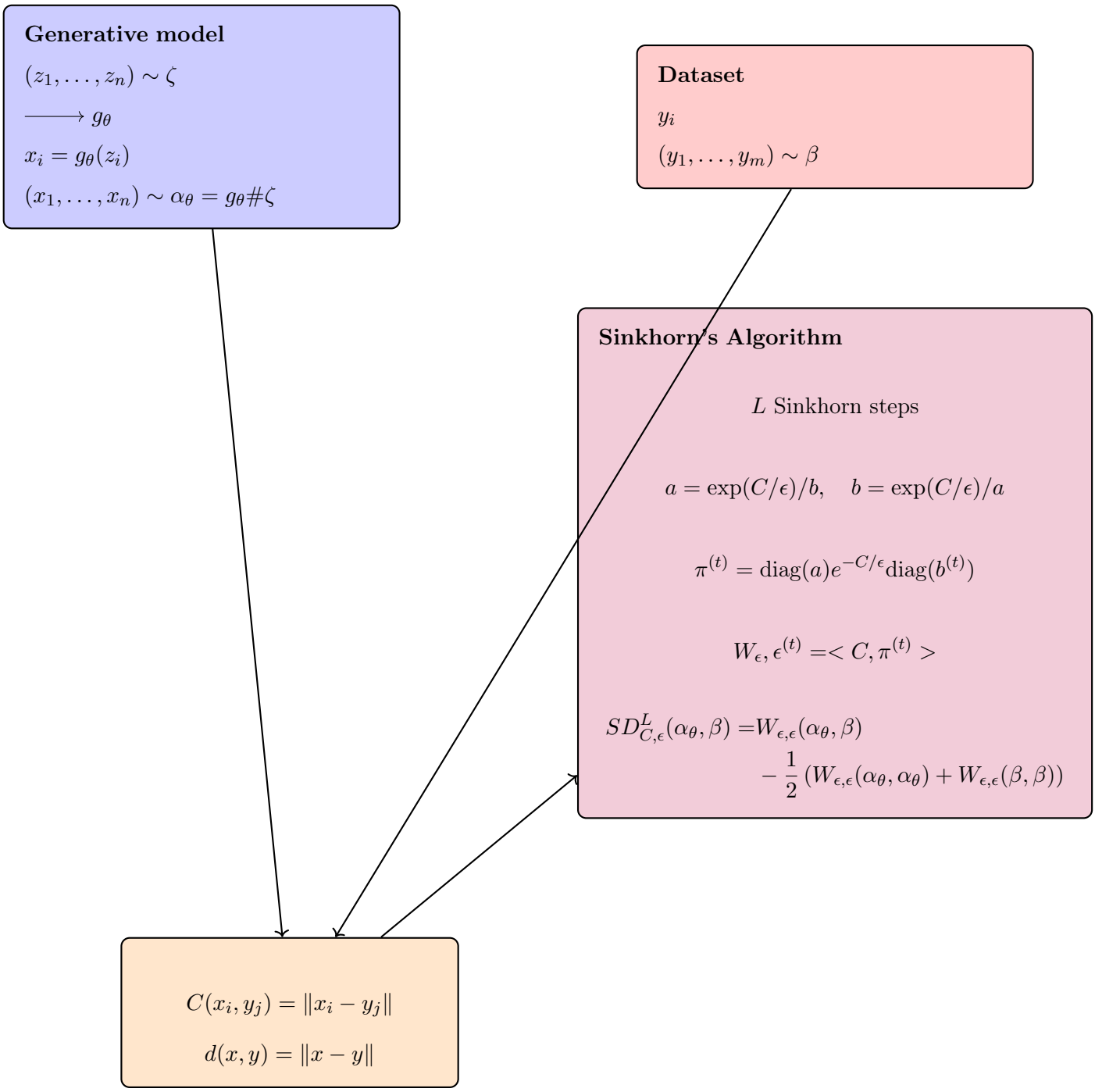

Figure 10 – Flow diagram for computing the Sinkhorn divergence estimated from samples $SD^L_{C,\epsilon}(\hat{\alpha}m, \hat{\beta}n)$. Samples from the generative model $\alpha\theta$ are obtained by applying the function $g\theta$ to samples from the initial measure ζ (blue block). These samples are combined with real data (red block) to compute a distance matrix C, used in the Sinkhorn iterations. The resulting loss is the one on which automatic differentiation is applied for parameter learning. The diagram shows a 2-layer neural network $g_\theta : z \mapsto x$, applicable to any generative model.

4. Algorithmic Foundations

A. Complementarity of the Primal and Dual Problems

Proposition 4.A.1
Let P^* and (f^*, g^*) be optimal solutions for the primal and dual problems respectively. Then, for any pair $(i, j) \in [n] \times [m]$, the following equality holds :

$$P^*_{i,j}(C_{i,j} - f^*_i + g^*_j) = 0.$$

Proof :

By strong duality :

$$\langle P^*, C\rangle = \langle f^*, a\rangle + \langle g^*, b\rangle.$$

$P^* 1_m = a$ and $P^{*T} 1_n = b$ therefore :

$$\langle f^*, a\rangle + \langle g^*, b\rangle = \langle f^*, P^* 1_m\rangle + \langle g^*, P^{*T} 1_n\rangle,$$

by the property of the Frobenius inner product

$$\langle f^*, P^* 1_m\rangle + \langle g^*, P^{*T} 1_n\rangle = \langle f^* 1_m^T, P^*\rangle + \langle 1_n g^{*T}, P^*\rangle,$$

therefore :

$$\langle P^*, C - f^* \oplus g^*\rangle = 0.$$

$C - f^* \oplus g^*$ is positive since (f^*, g^*) is a solution to the dual problem. Therefore $\forall (i, j)$ such that $P_{i,j} > 0$,

$$C_{i,j} - (f_i + g_j) = 0$$

i.e.

$$P^*_{i,j}(C_{i,j} - f^*_i + g^*_j) = 0.$$

Proposition 4.A.2
A matrix $P \in \mathbb{R}^{n\times m}$ and a pair of vectors (f, g) are complementary with respect to C if, for any pair of indices (i, j) such that $P_{i,j} > 0$, we also have $C_{i,j} = f_i + g_j$.

If P and (f, g) are feasible and complementary solutions for the primal and dual problems respectively, then P and (f, g) are optimal for both the primal and the dual.
Proof :
By weak duality :

$$L_C(a, b) \leq \langle P, C\rangle = \langle P, f \oplus g\rangle = \langle a, f\rangle + \langle b, g\rangle \leq L_C(a, b).$$

Thus, P and (f, g) are respectively optimal.

B. Graphs and Algorithms

Recall 4.B.1.
A vertex or extremal point of a convex set is defined as a point $\mathbf{x}$ such that, if $\mathbf{x} = (\mathbf{y} + \mathbf{z})/2$ for points $\mathbf{y}$ and $\mathbf{z}$ in that set, then $\mathbf{x} = \mathbf{y} = \mathbf{z}$.

With a linear program and a non-empty bounded solution set, the minimum is attained at a vertex or extremal point of that set [4]. The solution set $U(a, b)$ of the primal optimal transport problem is bounded and non-empty (see NW method below), so the search for an optimal solution P can be restricted to the extremal points of $U(a, b)$.

Let $V = \{1, 2, \ldots, n\}$ and $V' = \{1', 2', \ldots, m'\}$; the union of these sets, $V \cup V'$, contains $n+m$ nodes. The arcs between these nodes are represented by the set $\mathcal{E}$, containing all pairs (i, j') with $i \in [n]$ and

$j \in [m]$. Each edge (i, j') is associated with a cost C_{ij}. The complete bipartite graph G between V and V' is denoted $(V \cup V', \mathcal{E})$.

North-West Corner Rule (NW rule)
The north-west corner rule is an algorithm that constructs a vertex of $U(a, b)$ in at most $n + m$ operations. It can be used to initialize an algorithm such as the simplex algorithm, which we will discuss later.

Algorithm 2 North-West Corner Rule Algorithm

```
0: Initialize i ← 1, j ← 1, r ← a_1, c ← b_1
0: while i ≤ n and j ≤ m do
0:     t ← min(r, c)
0:     P_{i,j} ← t
0:     r ← r − t
0:     c ← c − t
0:     if r = 0 then
0:         increment i and update r ← a_i if i ≤ n
0:     end if
0:     if c = 0 then
0:         increment j and update c ← b_j if j ≤ m
0:     end if
0: end while=0
```

It is possible to obtain more solutions of this type by permuting a and b, then applying the algorithm and finally re-permuting the matrix to obtain a solution for a and b. The set of solutions thus obtained is called $\mathcal{N}(a, b)$. This set is included in the set of extremal points of $U(a, b)$ [5].

Proposition 4.B.2. (Extremal Solutions) Let P be an extremal point of the polytope $U(a, b)$. Let $\mathcal{S}(P)$ be the set of edges $\{(i, j')\}, i \in [n], j \in [m]$ such that $P_{ij} > 0$. Then, the graph $G(P) \triangleq (V \cup V', \mathcal{S}(P))$ has no cycles. In particular, P can have at most $n + m - 1$ non-zero values. [5]

Simplex Algorithm :

Part 1 : Idea of the Algorithm The idea of the simplex algorithm is to take an extremal solution $P \in U(a, b)$, then find an associated dual solution and check whether this solution is feasible. If so, P is optimal (see Part A). Otherwise, P must be perturbed to obtain a new extremal matrix until an optimal solution is found.

Part 2 : Finding a Dual Solution Complementary to a Primal Solution

This is done simply by finding two vectors f and g such that, for any edge (i, j') in $\mathcal{S}(P)$, $f_i + g_j = C_{ij}$.

Let s be the cardinality of $\mathcal{S}(P)$. Since P is extremal, $s \leq n + m - 1$. Finding f and g therefore requires solving the following system of s linear constraints in $n + m$ variables :

$$\begin{cases} f_{i_1} + g_{j_1} = C_{i_1 j_1} \\ f_{i_2} + g_{j_2} = C_{i_2 j_2} \\ \vdots \\ f_{i_s} + g_{j_s} = C_{i_s j_s} \end{cases} \tag{76}$$

$(i_1, j_1), (i_2, j_2), ... = \mathcal{S}(P)$

Since $s \leq n + m - 1 < n + m$, the linear system (76) is always underdetermined. However, $G(P)$ has no cycles, so $G(P)$ is either a tree or a forest (a union of trees). Thus, a simple graph traversal initializing the root to 0 allows us to obtain f and g as a complementary solution to P.

Part 3 : Simplex Algorithm and Graph Update

Algorithm 3 Simplex Algorithm

```
0:
0: procedure NetworkSimplex(P, C)
0:    Compute (f, g) complementary to P
0:    while there exists an edge (i, j) such that f_i + g_j > C_ij do
0:       Add edge (i, j) to the graph G(P)
0:       if G(P) is a forest then
0:          Use the simplex algorithm on the new graph
0:       else{G(P) contains a cycle}
0:          Use the Simplex on the update of G(P) performed with algorithm 4
0:       end if
0:    end while
0: end procedure=0
```

Algorithm 4 $G(P)$ Update Algorithm

0: Identify and note the cycle $(i_1, j_1), (j_1, i_2), (i_2, j_2), \ldots, (i_l, j_l), (j_l, i_1)$.
0: Compute θ, the largest possible increase at index (i, j) using this cycle :

$$\theta = \min_{k \leq l} P_{i_k, j_k}$$

0: **for** each $k \leq l$ **do**
0: **if** (i_k, j_k) is a positive edge **then**
0: $P_{i_k, j_k} \leftarrow P_{i_k, j_k} + \theta$
0: **else**
0: $P_{i_k, j_k} \leftarrow P_{i_k, j_k} - \theta$
0: **end if**
0: **end for**
0: Remove edge (i_{l+1}, j_k) from $G(P)$.
0:
0: =0

5. Conclusion

In this study, we have succeeded in providing the first keys to understanding the field of optimal transport. We have explored the fundamental foundations and provided initial answers to the problems encountered. However, it is important to note that our exploration only scratches the surface of the subject. The field of optimal transport is vast and complex, and there is still much to learn and discover.

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